

```
[vbg@vbg-work ~]$
```

Surprisingly, our ACL did not work. Despite the fact that we specified any/all users to write to the directory, we were unable to do so. To enable anonymous writes (not at all recommended) to your directory, we will have to use another security directive—*allow*.

To enable anonymous updates to directory, add the following global configuration option to *slapd.conf*:

```
allow update_anon
```

Save the configuration file, check for errors in the configuration file, restart the *ldap* service and test whether you can add a user anonymously. This should succeed.

The *allow update_anon* option generally should not be enabled, as we do not want anyone to write data into our directory without authentication. It has some very special use cases that we might cover in the later part of this series.

Now, let us apply a more common security requirement to our LDAP server—disable anonymous access altogether. In other words, valid authentication will be required even for the *ldapsearch* command. How would we create that ACL?

```
access to * by anonymous none
```

The above ACL does not give any permissions to anonymous users. You can check whether the syntax of the ACL is correct or not, using the *slaptest* command discussed earlier.

But what are the possible permissions that can be specified in *slapd.conf*? We have already seen three of them: *read*, *write* and *none*. The others are: *discard*, *auth*, *compare* and *search*. We will go into the details of each of these permissions later; we intuitively know the meaning of the other permissions—*read*, *write* and *none*.

Similarly, what are the possible values for 'who'? We have discussed two of these: *** (*all*) and *anonymous*. Two other possible values are: *users* and

self. *users* means entries or DNs that exist in the directory, and *self* means the user who is currently authenticated.

There are many possible values for 'what'. This depends on the directory data. In this article, we have only discussed *** as a possible value for 'what'. As we delve deeper into LDAP and understand attributes, objectclasses and schema, we will be able to better define the 'what' part of an ACL.

ACLs can also take more complex forms, including a 'control' option as per the security requirements of a directory. We will touch on these aspects once we create a central directory with users and passwords for authentication. For those of you who would like to experiment with ACLs right away and not wait for the upcoming articles, the man page for *slapd.access* is a storehouse of information:

```
[root@vbg ~]# man slapd.access
```

This page contains a wealth of information on *slapd.conf* access control options and is a must read when securing an LDAP server.

In the next article in this series, we will explore LDAP concepts such as schema, objectclasses and attributes, and build an LDAP server capable of authenticating Linux users. Do drop me a mail if you have any doubts or questions. The first article has elicited quite a few interesting responses—from questions to requests for certain advanced topics. I hope that we'll be able to cover most of them in the ongoing series. 

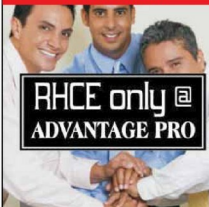
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